



Adyapan School

Structural Analysis



Duration - 3 months

**Industry
Certification**



Skill India Certified

**250+
Partner Companies**

Master the backbone of every **structure**

Analyze smarter, design safer structures.

This 12-week career-focused Structural Analysis course helps you bridge the gap between classroom concepts and engineering practice. With applied learning, numerical problem solving, and real case discussions, you will gain expertise in structural behavior, load distribution, and system analysis—preparing you for real-world engineering challenges.

12

WEEKS

30+

PROGRAMS OFFERED

20,000+

STUDENTS

250+ PARTNERED COMPANIES



ABOUT ADYAPAN SCHOOLS

Where education meets real-world impact

Not just a course — a platform to launch
your career.

Adyapan Schools was built with a single conviction:
learning works best when it happens in the real world.
We partner with top companies, mentors, and industry
platforms to ensure every student graduates with a
portfolio of work that speaks louder than a certificate.

Our programs combine rigorous coursework with live
client projects, giving you the skills and proof-of-work
that employers actually want.

MISSION

To equip ambitious learners with
practitioner-level digital
marketing skills through mentor-
led, project-based education that
bridges the gap between learning
and earning.



VISION

To be India's most trusted
launchpad for the next generation
of marketing leaders — defined
not by degrees but by the real
work.



Everything you need to grow fast

PROGRAM HIGHLIGHTS

Live Industry Projects

Work on campaigns for real brands alongside your coursework. Build portfolio projects that prove your expertise to employers.

1-on-1 Mentorship

Dedicated mentors from Google, Microsoft, Mastercard and more. Get personalized guidance and industry connections.



AI-Powered Marketing

Learn cutting-edge AI tools alongside evergreen fundamentals. Stay ahead of the curve in a rapidly evolving landscape.



Dual Certification

Earn both a Course Completion and Internship Certificate – accredited by Skill India Digital Hub and NSDC.



Internship Guarantee

Graduate with an internship completion certificate from a live brand project. Concrete, resume-ready proof of work.



Industry Network

Join a network of alumni at Amazon, Google, Adobe, Microsoft. Access exclusive hiring events and referral opportunities.

12 weeks. 12 modules. Infinite impact.

WEEK 1

Basics of Structural Analysis, Types of Structures, Loads, and Supports

- Discussion of Curriculum
- Introduction to structural systems: beams, trusses, frames, cables
- Introduction to beams and loading conditions
- Classification: determinate vs indeterminate structures
- Types of loads: point load, UDL, UVL, moment loads
- Types of supports: roller, hinge, fixed (with reaction components)
- Real-life examples: bridges, buildings, towers



WEEK 2

Determinate Structures, Equilibrium Equations, Free Body Diagrams

- Concept of static determinacy and stability
- Equilibrium equations: $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$
- Drawing accurate Free Body Diagrams (FBDs)
- Support reactions calculation
- Common mistakes in FBDs and how to avoid them



WEEK 3

Trusses - Method of Joints

- Difference between joints vs sections method
- Cutting sections strategically
- Solving for internal forces directly
- Efficiency in solving large trusses
- Mixed problems (joints + sections)



12 weeks. 12 modules. Infinite impact.

WEEK 4

Trusses – Method of Sections

- Difference between joints vs sections method
- Cutting sections strategically
- Solving for internal forces directly
- Efficiency in solving large trusses
- Mixed problems (joints + sections)



WEEK 5

Beams – Shear Force and Bending Moment

- Shear force and bending moment concepts
- Sign conventions (very important for exams)
- The relation between load, shear force, and bending moment
- Drawing basic SFD & BMD



WEEK 6

Beams – SFD & BMD Problems

- Advanced SFD and BMD for different loading cases
- UDL, UVL, and point load combinations
- Maximum bending moment and points of contraflexure
- Practice numerical problems
- Interpretation of diagrams (real-world relevance)



12 weeks. 12 modules. Infinite impact.

WEEK 7

Frames and Machines Basics

- Difference between frames and trusses
- Internal force analysis in frames
- Machines vs structures (force vs motion)
- Free body diagrams for frames
- Simple frame problem solving



WEEK 8

Deflection of Beams - Double Integration Method

- Concept of deflection and slope
- Differential equation of elastic curve
- Step-by-step double integration method
- Boundary conditions application
- Solving standard beam deflection problems



WEEK 9

Deflection - Macaulay's Method

- Need for Macaulay's method (handling multiple loads)
- Macaulay's notation and step functions
- Solving complex loading cases
- Comparison with the double integration method
- Practice numerical problems



12 weeks. 12 modules. Infinite impact.

WEEK 10

Indeterminate Structures Introduction

- Difference: determinate vs indeterminate structures
- Degree of indeterminacy calculation
- Compatibility conditions
- Basic concepts of redundancy
- Real-world applications (continuous beams, fixed structures)



WEEK 11

Moment Distribution Method

- Concept of stiffness and distribution factors
- Carry-over factor explanation
- Step-by-step moment distribution table
- Solving continuous beam problems
- Advantages vs other methods



WEEK 12

Revision, Case Studies, Mini Project

- Revision of all core topics (trusses, beams, deflection)
- Case studies: bridges, buildings (failure/success analysis)
- Mini Project:
 - Analyze a real structure (beam/truss)
 - Draw SFD, BMD, or force diagram
 - Final assessment + conceptual clarity session



WHO THIS IS FOR

This course is perfect for

Students & Career Switchers

Aspiring Structural Engineers

Design Engineers

Site & Construction
Engineers

Civil Engineering Students

Forensic Engineers

CERTIFICATIONS



ALUMNI NETWORK

Our alumni work at world-class companies

Amazon

Adobe

Google

Autodesk

Microsoft

Deloitte

Your career switch is one click away.

Ready to begin? Apply at adyapanschool.com or email us at support@adyapan.com

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